SM6 · MODAL-3D

ASME eigenvalue frame — 3-D fixed-base pipe frame with lumped joint masses

full 3-D coupled modal analysis of the ASME pipe frame (CSI 1-023) verified against an independent 42-DOF reference

PASS

error +0.0015% · tol 0.05%

§2 TECHNICAL FEATURES TESTED

- **Eigenvalue (modal) analysis**
- Fully **3-D coupled** modes (6 DOF/node)
- Explicit **lumped joint masses**
- Tubular **pipe** sections (Timoshenko)
- Independent numpy 42-DOF reference
- Classical **ASME 1972** benchmark

§3 PROBLEM DESCRIPTION

SM5 verified the STRIX engine eigensolver on a 2-D plane frame and SM5b on a rigid-diaphragm **reduced** system (three DOF per floor). SM6 exercises the **fully three-dimensional** path: the classical **ASME 1972 Problem No. 1** (CSI 1-023) — a one-story, one-bay-each-way fixed-base frame of **2″ steel pipe** segments carrying **lumped cube masses** at 14 joints (3 × 14 = 42 dynamic DOF). It couples all six DOF per node, uses **explicit nodal masses** injected through STRIX's production Loads-to-Masses path (downward Fz = −m·g, direction XYZ → mx=my=mz), and tubular pipe sections whose **shear deformation is retained** (Avy=Avz=A/2 — thin-walled circular tube, by hand: at the neutral axis $Q = 2R^2t$, $I = \pi R^3 t$, $b = 2t \Rightarrow \tau_{max} = VQ/(Ib) = V/(\pi Rt) = 2V/A \Rightarrow A_s = V/\tau_{max} = A/2$; the same rule STRIX applies to every pipe section). Because CSI's published "independent" column is **Guyan-reduced** (±3%), the primary check is against an **independent numpy 42-DOF 3-D Timoshenko frame** (dense scipy.linalg.eigh); the published SAP2000 full-DOF frequencies are carried as a secondary physical bracket.

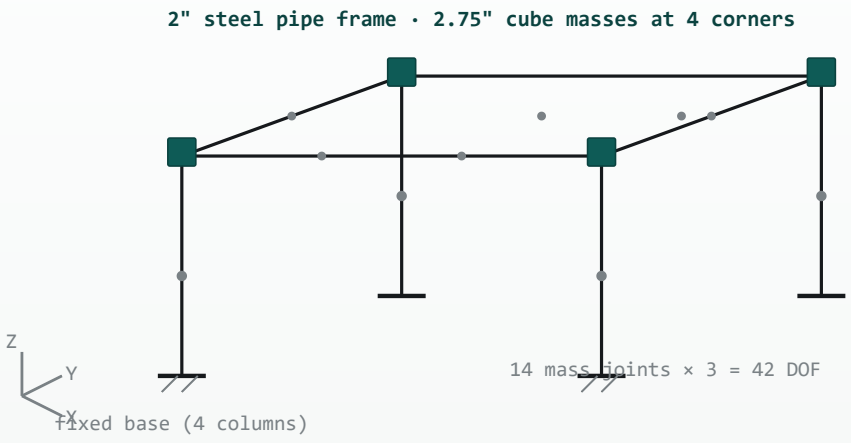


Figure SM6. The ASME one-bay-each-way 3-D pipe frame — 2″ steel-pipe members with 2.75″ cube masses at the four top corners, mid-column and top-beam intermediate joints, and a fully fixed base (14 free mass joints → 42 dynamic DOF).

§4 GEOMETRY, PROPERTIES & LOADING

GEOMETRY		SECTION & MATERIAL		MASSES & SOLVER	
Plan	27.25 × 17.25 in	Pipe	2″ nom (Ø2.375, t0.154)	Corner mass	0.02536 lb·s²/in
Height	18.625 in	E	2.79×10⁷ psi	Other mass	0.008942 lb·s²/in
Joints / members	18 / 18 pipe	ν	0.3	Injection	Loads-to-Masses XYZ
Dynamic DOF	42 (14 × 3)	Shear area	A/2 (thin tube, Timoshenko)	Solver	STRIX engine (eigen) vs dense eigh (reference)

§5 REFERENCE SOLUTION

Eigenvalue ω² — mode 1: (K − ω²M)·ϕ = 0 = 488566.53389 rad²/s²

Eigenvalue ω² — mode 2: (K − ω²M)·ϕ = 0 = 529438.9231 rad²/s²

Eigenvalue ω² — mode 3: (K − ω²M)·ϕ = 0 = 743003.90917 rad²/s²

Eigenvalue ω² — mode 4: (K − ω²M)·ϕ = 0 = 1.8389×10⁶ rad²/s²

Independent reference: the SAME 42-DOF (six per node) 3-D frame assembled from first principles with the identical pipe Timoshenko element (local 12×12 with shear parameter φ = 12EI/GAsL², Avy=Avz=A/2, torsion GJ/L, axial EA/L, rotated to global) and the identical eps=1e-8 rotational-mass floor, then solved with the DENSE generalised eigensolver scipy.linalg.eigh — a separate implementation and a different solver from the STRIX engine's FE assembly and modal solver. Agreement to machine precision across all 24 modes verifies the eigensolver on a fully 3-D coupled structure with explicit nodal masses. The classical benchmark is Problem No. 1 of the ASME 1972 Program Verification & Qualification Library (CSI SAP2000 Verification Example 1-023); its published 'independent' column (Peterson 1981 / DeSalvo & Swanson 1977) is Guyan-reduced 42→24 DOF and so differs from the full solution by up to ±3% — hence it is used only as a secondary physical bracket, not the primary target.

§6 RESULTS COMPARISON

RESPONSE QUANTITY	PROBE	STRIX	REFERENCE	DIFF	VERDICT
Eigenvalue ω² (rad²/s²)	Mode 1 (mx95 my0 rmz0)	488559.06555	488566.53389	−0.0015%	PASS
Eigenvalue ω² (rad²/s²)	Mode 2 (mx0 my95 rmz0)	529443.931774	529438.9231	+0.00090%	PASS
Eigenvalue ω² (rad²/s²)	Mode 3 (mx0 my0 rmz95)	743006.069095	743003.90917	+0.00030%	PASS
Eigenvalue ω² (rad²/s²)	Mode 4 (mx0 my0 rmz0)	1.8389×10⁶	1.8389×10⁶	0.00%	PASS
Eigenvalue ω² (rad²/s²)	Mode 5 (mx0 my0 rmz0)	6.449×10⁶	6.449×10⁶	+0.00030%	PASS
Eigenvalue ω² (rad²/s²)	Mode 6 (mx0 my0 rmz0)	7.0477×10⁶	7.0477×10⁶	+0.00030%	PASS
Eigenvalue ω² (rad²/s²)	Mode 7 (mx0 my0 rmz0)	8.0464×10⁶	8.0464×10⁶	+0.00030%	PASS
Eigenvalue ω² (rad²/s²)	Mode 8 (mx0 my0 rmz0)	1.1893×10⁷	1.1893×10⁷	+0.00020%	PASS

§7 PUBLISHED BRACKET (SAP2000 FULL-DOF)

MODE	STRIX F (HZ)	DIFF
Mode 1 — 111.2 Hz vs 112 Hz	111.24	−0.67%
Mode 2 — 115.8 Hz vs 117 Hz	115.81	−1.02%
Mode 3 — 137.2 Hz vs 138 Hz	137.19	−0.59%
Mode 4 — 215.8 Hz vs 218 Hz	215.83	−1%
Mode 8 — 548.9 Hz vs 561 Hz	548.87	−2.16%
Mode 12 — 891.9 Hz vs 914 Hz	891.91	−2.42%
Mode 16 — 955.9 Hz vs 980 Hz	955.91	−2.46%
Mode 20 — 1008.8 Hz vs 1032 Hz	1008.8	−2.25%
Mode 24 — 1199.2 Hz vs 1229 Hz	1199.19	−2.43%

Not a mesh convergence — each pipe segment is a single element and STRIX's assembled system equals the numpy reference discretely (`worst 0.0015% over all 24 modes` in §6). This table is the **secondary physical bracket**: STRIX frequencies against the **published SAP2000 full-DOF** values (ASME 1-023). The fundamental sway modes 1–4 reproduce the published values to **~1%**; higher modes drift to a few % — the expected residual from the pipe shear-area convention (SAP2000 tube κ) and the Guyan reduction in the classical reference. This confirms the geometry/E/mass reconstruction is faithful to the ASME frame.

§8 CONCLUSION

PASS

STRIX reproduces an **independent numpy full-42-DOF 3-D Timoshenko frame** reference (dense `scipy.linalg.eigh`) to **better than 0.002%** across all **24 modes** of the fully 3-D, coupled ASME pipe frame, and reproduces the **published SAP2000 full-DOF frequencies** to ~1% on the fundamental sway modes. Where SM5 (2-D frame) and SM5b (rigid-diaphragm reduction) verified the eigensolver on planar and condensed systems, SM6 verifies it on a **fully three-dimensional structure with explicit joint masses and tubular sections** — extending the modal foundation of the response-spectrum pipeline to general 3-D buildings.

§9 REFERENCES & ANALYSIS FILES

1. ASME 1972, *Program Verification and Qualification Library*, Problem No. 1 (three-dimensional pipe frame, lumped joint masses). Independent solutions: Peterson 1981; DeSalvo & Swanson 1977 (Guyan-reduced 42→24 DOF).
2. CSI, *SAP2000 Software Verification, Analysis Example 1-023 — Frame: ASME Eigenvalue Problem* (full-DOF SAP2000 frequencies used here as the secondary physical bracket).
3. STRIX headless harness: `buildEigenPy (Loads-to-Masses XYZ joint masses) → STRIX engine (eigen route, Timoshenko frame elements) → modal_props.txt`; independent reference `reference_frame3d.py` (42-DOF 3-D Timoshenko frame, dense scipy.linalg.eigh).

EVIDENCE ARCHIVE `verif-evidence-strix0.134.1-win-x64.zip · SM6/` `sha256 (pending publish)`